CS246: Database Management Systems Lab

Lab # 10 (1 Questions, 135 Marks)
Lab session: AL1
Held on: 17-Mar-2024 (Mon)
Lab Timings: 14:00 to 17:00 Hours Pages: 5
Submission time: 16:45 Hrs, 17-Mar-2024
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1. This lab theme is centered around section 5.3 Triggers of the text book *Database System Concepts* Abraham Silberschatz, Henry F Korth & S. Sudarshan.
2. Manual pages for trigger are attached.
3. Tutorial slides on triggers are available in ”Files” tab.
4. Some of the tasks are so designed to have errors. You should not remove them. Instead demonstrate the error as it is.

Question 1: (135 points)
Using `MySQL` perform the following tasks:

Task 01 (1 mark) Create a database named *week10*. Create the following tables inside this database.

Task 02 (3 marks) Perform the following tasks.

1. (1 mark) Create `sailors` table with the following description

column #	name	data type	constraint
1	`sid`	`int`	primary key
2	`sname`	`char(50)`	None
3	`rating`	`int`	None
4	`age`	`decimal(3,1)`	None

2. (1 mark) Create `boats` table with the following description

column #	name	data type	constraint
1	`bid`	`int`	primary key
2	`bname`	`char(50)`	None
3	`bcolor`	`char(50)`	None

3. (1 mark) Create `reserves` table with the following description

column #	name	data type	constraint
1	sid	int	refers to sailors
2	bid	int	refers to boats
3	day	char(50)	None
			sid, bid, day is primary key
			sid foreign key cascade on update
			sid foreign key cascade on delete
			bid foreign key cascade on update
			bid foreign key cascade on delete

Task 03 (5 marks) Create the following log tables as per the specification given below

1. (1 mark) sailors_log

column #	name	data type	values
1	sid	int	
2	event_ba	char(50)	{before, after}
3	ops	char(50)	{insert, update, delete}
4	date_time	datetime	system time

2. (1 mark) boats_log

column #	name	data type	values
1	bid	int	
2	event_ba	char(50)	{before, after}
3	ops	char(50)	{insert, update, delete}
4	date_time	datetime	system time

3. (1 mark) reserves_log

column #	name	data type	values
1	sid	int	
2	bid	int	
3	day	char(10)	
4	event_ba	char(50)	{before, after}
5	ops	char(50)	{insert, update, delete}
6	date_time	datetime	system time

4. (1 mark) sailors_log_log

column #	name	data type	values
1	sid	int	
2	event_ba	char(50)	{before, after}
3	ops	char(50)	{insert, update, delete}
4	date_time	datetime	system time

5. (1 mark) sailors_log_log_log

column #	name	data type	values
1	sid	int	
2	event_ba	char(50)	{before, after}
3	ops	char(50)	{insert, update, delete}
4	date_time	datetime	system time

Task 04 (3 marks) Populate data

1. (1 mark) populate the `sailors` table from the file `sailors01.csv`
2. (1 mark) populate the `boats` table from the file `boats01.csv`
3. (1 mark) populate the `reserves` table from the file `reserves01.csv`

Task 05 (45 marks) Create the following triggers

1. (5 marks) Create a trigger on `sailors` table say `sailor_t1`. Whenever a row is inserted into this table, and before the row is inserted this trigger should insert a record into `sailor_log` table with `sid` of the row about to be inserted, `event_ba="before"`, `ops="insert"`
2. (5 marks) Create a trigger on `boats` table. Whenever a row is inserted into this table, and before the row is inserted this trigger should insert a record into `boats_log` table with `bid` of the row about to be inserted, `event_ba="before"`, `ops="insert"`
3. (5 marks) Create a trigger on `reserves` table. Whenever a row is inserted into this table, and before the row is inserted this trigger should insert a record into `reserves_log` table with `sid, bid, day` of the row about to be inserted, `event_ba="before"`, `ops="insert"`

4. (5 marks) Create a trigger on `sailors` table. Whenever a row is updated, and after the row is updated this trigger should insert a record into `sailor_log` table with `sid` of the row about to be updated, `event_ba="after"`, `ops="update"`
5. (5 marks) Create a trigger on `boats` table. Whenever a row is updated, and after the row is updated this trigger should insert a record into `boats_log` table with `bid` of the row about to be updated, `event_ba="after"`, `ops="update"`
6. (5 marks) Create a trigger on `reserves` table. Whenever a row is updated, and after the row is updated this trigger should insert a record into `reserves_log` table with updated row inserted into this table and `event_ba="after"`, `ops="update"`

7. (5 marks) Create a trigger on `sailors` table. Whenever a row is deleted, and after the row is deleted this trigger should insert a record into `sailor_log` table with `sid` of the deleted row, `event_ba="after"`, `ops="deleted"`
8. (5 marks) Create a trigger on `boats` table. Whenever a row is deleted, and after the row is deleted this trigger should insert a record into `boats_log` table with `bid` of the deleted row, `event_ba="after"`, `ops="deleted"`
9. (5 marks) Create a trigger on `reserves` table. Whenever a row is deleted, and after the row is deleted this trigger should insert a record into `reserves_log` table with `bid` of the deleted row, `event_ba="after"`, `ops="deleted"`

Task 06 (27 marks) populate data and show the log files

1. (1 mark) populate the `sailors` table from the file `insert-sailors02.csv`
2. (1 mark) populate the `boats` table from the file `insert-boats02.csv`
3. (10 mark) populate the `reserves` table by

(a) randomly generating a `sid` from `insert-sailors02.csv` entries
(b) randomly generating a `bid` from `insert-boats02.csv` entries
(c) randomly generating a `day` between `2024-01-01` and `2024-12-31`
(d) Insert this record into `reserves` table
(e) Continue insert until every sailor is reserved each bid two times

4. (1 mark) list the contents of `sailors_log` table
5. (1 mark) list the contents of `boats_log` table
6. (1 mark) list the contents of `reserves_log` table

7. (1 mark) Update the `sailors` table `rating` from the file `update-sailors02.csv`
8. (1 mark) Update the `boats` table `bcolor` from the file `update-boats02.csv`
9. (1 mark) Update the first 100 entries of `reserves` table by incrementing the date by 1 day.
10. (1 mark) list the contents of `sailors_log` table
11. (1 mark) list the contents of `boats_log` table
12. (1 mark) list the contents of `reserves_log` table

13. (1 mark) Delete the records from the `sailors` table whose `sid`s are given in `delete-sailors02.csv`
14. (1 mark) Delete the records from the `boats` table whose `bid`s are given in `delete-boats02.csv`
15. (1 mark) Delete the first 100 records from the `reserves` table.

16. (1 mark) list the contents of `sailors_log` table
17. (1 mark) list the contents of `boats_log` table
18. (1 mark) list the contents of `reserves_log` table

Task 07 (10 marks) Multiple triggers

1. (5 marks) Create a trigger on `sailors` table say `sailor_t2`. Whenever a row is inserted into this table, and before the row is inserted this trigger should insert a record into `sailor_log` table with `sid` of the row about to be inserted, `event_ba="before"`, `ops="t2 insert"`. `sailor_t2` trigger should execute before `sailor_t1` trigger.
2. (5 marks) Create a trigger on `sailors` table say `sailor_t3`. Whenever a row is inserted into this table, and before the row is inserted this trigger should insert a record into `sailor_log` table with `sid` of the row about to be inserted, `event_ba="before"`, `ops="t3 insert"`. `sailor_t3` trigger should execute before `sailor_t1` trigger and after `sailor_t2`.

Task 08 (2 marks) populate data and show the log files

1. (1 mark) populate the `sailors` table from the file `sailors03.csv`
2. (1 mark) list the contents of `sailors_log` table

Task 09 (10 marks) Recursive triggers

1. (5 marks) Create a trigger on `sailors_log` table say `sailor_log_t1`. Whenever a row is inserted into this table, and after the row is inserted this trigger should insert a record into `sailor_log_log` table with appropriate entries into the log file.
2. (5 marks) Create a trigger on `sailors_log_log` table say `sailor_log_log_t1`. Whenever a row is inserted into this table, and after the row is inserted this trigger should insert a record into `sailor_log_log_log` table with appropriate entries into the log file.

Task 10 (4 marks) populate data and show the log files

1. (1 mark) populate the `sailors` table from the file `sailors04.csv`
2. (1 mark) list the contents of `sailors_log` table
3. (1 mark) list the contents of `sailors_log_log` table
4. (1 mark) list the contents of `sailors_log_log_log` table

Task 11 (5 marks) Declare the following

1. (5 marks) Create a trigger on `sailors_log_log_log` table say `sailor_log_log_log_t1`. Whenever a row is inserted into this table, and after the row is inserted this trigger should insert a record into `sailors` table with randomly generated entries.

Task 12 (4 marks) populate data and show the log files

1. (1 mark) populate the `sailors` table from the file `sailors05.csv`
2. (1 mark) list the contents of `sailors_log` table
3. (1 mark) list the contents of `sailors_log_log` table
4. (1 mark) list the contents of `sailors_log_log_log` table

Task 13 (6 marks) populate data and show the log files

1. (1 mark) insert 10 duplicate records into `sailors` table
2. (1 mark) insert 10 duplicate records into `boats` table
3. (1 mark) insert 10 duplicate records into `reserves` table

4. (1 marks) show the contents of `salors_log`
5. (1 marks) show the contents of `boats_log`
6. (1 marks) show the contents of `reserves_log`